Covariance, Correlation, and Regression Slope – Quick Reference

1. Covariance vs. Correlation

- Covariance (Cov(X, Y)):
  Measures the raw direction and magnitude of joint variation between X and Y.
  Units: product of X and Y units.
  Formula: $Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

- Correlation (ρ):
  Standardized measure of linear relationship between X and Y.
  Range: [-1, 1]. Unitless.
  Formula: $\rho = Cov(X, Y) / (\sigma_X * \sigma_Y)$

2. Regression Slope (β_1)

- In linear regression ($Y = \beta_0 + \beta_1 X$), the slope β_1 is computed as:
  $\beta_1 = Cov(X, Y) / Var(X)$

- Alternatively, using correlation:
  $\beta_1 = \rho * (\sigma_Y / \sigma_X)$

This shows that the slope depends on both the correlation and the scale (standard deviations) of X and Y.

3. Interpretation on an XY Scatter Plot

- Covariance tells whether X and Y move together, but not how strong the relationship is relative to scale.
- Correlation tells how closely the points follow a straight line.
- The regression slope indicates how much Y increases (on average) when X increases by 1 unit.

4. Example

Let:
  Cov(X, Y) = 2.0
  Var(X) = 1.0
Then:
  $\beta_1 = 2.0$

But if:
  Var(X) = 4.0
Then:
  $\beta_1 = 2.0 / 4.0 = 0.5$ (flatter slope due to more spread in X)

Also:
If $\rho = 0.8$, $\sigma\_X = 1.5$, $\sigma\_Y = 3.0$
Then:
$Cov(X, Y) = \rho * \sigma\_X * \sigma\_Y = 0.8 * 1.5 * 3.0 = 3.6$
$\beta_1 = \rho * (\sigma\_Y / \sigma\_X) = 0.8 * (3.0 / 1.5) = 1.6$

5. Summary Table

Concept	Formula	Description
Covariance	Cov(X, Y)	Raw joint movement
Correlation	$\rho = Cov(X,Y) / (\sigma\_X * \sigma\_Y)$	Strength of linear relation
Slope	$\beta_1 = Cov(X, Y) / Var(X)$	ΔY per unit ΔX
Slope (alt)	$\beta_1 = \rho * (\sigma\_Y / \sigma\_X)$	Uses correlation and scale

6. Common Names for the Slope

- Regression Coefficient: General term in linear regression.
- Slope of the Regression Line: Visual interpretation on a graph.
- Beta Coefficient (β): In finance, measures sensitivity to market.
- Least Squares Estimate of Slope: Specific to OLS method.
- Covariance Slope (informal): Occasionally used to emphasize its formula, but not standard.